

Fig. 6: Punch card

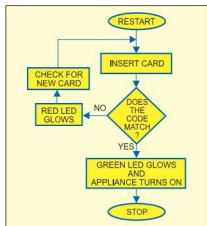


Fig. 7: Flow-chart of the program

circuit, the 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805 (IC4). Capac-

itor C2 bypasses the ripples present in the regulated supply. LED11 acts as the power indicator and R13 limits the current through LED1. Fig. 3 shows the pin details of regulator 7805 and BC337.

Construction

An actual-size, single-side PCB for the electronic card lock is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error. Use IC base for placing the microcontroller on the PCB. Check the supply voltage. Align the LDRs and LEDs such that these directly face each other. Cover the LDRs with suitable masking to protect these from other light sources. The card is shown in Fig. 6.

Insert the card into the card sensor slot, close switch S2 and then press switch S3. The internal LEDs of the optocoupler glow depending on the holes on the card. The internal transistors of the optocoupler drive into saturation and its collector goes low.

Software

The software for the electronic card lock is written in Assembly language

PARTS LIST

Semiconductors:

- IC1 - AT89C51 microcontroller
- IC2, IC3 - ILQ74 quad optocoupler
- IC4 - 7805, 5V regulator
- T1 - BC337 npn transistor
- LED1-D5 - 1N4007 rectifier diode
- LED1-LED8 - 5mm white LED
- LED9 - 5mm green LED
- LED10, LED11 - 5mm red LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R8, R10, R11 - 150-ohm
- R9 - 10-kilo-ohm
- R12 - 1-kilo-ohm
- R13 - 470-ohm
- RNW1 - 10-kilo-ohm resistor network

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 0.1 μ F ceramic disk
- C3 - 10 μ F, 16V electrolytic
- C4, C5 - 22 μ F ceramic disk

Miscellaneous:

- X1 - 230V, 500mA secondary transformer
- X_{AL} - 12MHz crystal
- LDR1-LDR8 - LDR
- RL1 - 12V, 1C/O relay
- S1, S3 - Push-to-on tactile switch
- S2 - On/off switch

and compiled using the Keil μ Vision4 compiler. The generated hex code is burnt into the microcontroller by using a suitable programmer. The program works as per the flow-chart shown in Fig. 7.

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LOCK.ASM

LOC	OBJ	LINE	SOURCE				
0000		1	ORG 0000h	007E	759003	19	MATCH: MOV P1, #00000011b
0000	020033	2	LJMP MAIN	0081	80FE	20	STAY: S JMP STAY
0013		3	ORG 0013h	0083	758901	22	DELAY: MOV TMOD, #01
0013	020070	4	LJMP INTF	0086	790F	23	MOV R1, #0FH
0030		5	ORG 0030h	0088	758B01	24	AGAIN: MOV TL1, #01H
0030	758160	6	MOV SP, #60H ; set stack pointer	008B	758D00	25	MOV TH1, #00h
0033	75A884	7	MAIN: MOV 16, #100001001b	008E	D28E	26	SETB TR1
0036	75A0FF	8	MOV P2, #0Fh	0090	308FFD	27	JNB TF1, BACK
0039	759000	9	MOV P1, #00h	0093	C28E	28	CLR TR1
003C	80FE	10	HERE: SJMP HERE	0095	C28F	29	CLR TF1
0070		11	ORG 0070h	0097	D9EF	30	DJNZ R1, AGAIN
0070	AAA0	12	INTP: MOV R2, P2	0099	22	31	ret
0072	BA0902	13	CJNE R2, #00001001b, nomatch	009A	7BFF	32	DELAY1: MOV R3, #0FFH
0075	8007	14	SJMP MATCH	009C	00	33	LOOP: NOP
0077	D292	15	NOMATCH: SETB pl.2	009D	00	34	NOP
0079	1183	16	ACALL DELAY	009E	DBFC	35	DJNZ R3, LOOP
007B	C292	17	CLR P1.2	00A0	22	36	RET
007D	32	18	RETI			37	END



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